The Earth also has two magnetic poles, north and south. Magnetic force lines depart from the south magnetic pole, swing around the Earth, and descend into the north magnetic pole (Figure 2.8). These force lines grab a compass needle, just as they grab iron filings, and drive the needle to point as nearly along the force lines as possible. That's how a compass works.



Fig. 2.7. The world's first commerical magnetically levitated train, in Shanghai, China.